Pathfinder Software Requirements Specification (SRS)

1. Scope

This SRS describes the software requirements for "Pathfinder," a web application developed using React, HTML, CSS, and Node.js, designed for high school and college students entering the workforce. The application provides users with comprehensive assistance through a personalized user interface, offering career guidance, job matching, resume analysis, and an AI-powered chatbot assistant.

2. Referenced Documents

- **API Documentation for OpenAI**
- **Firebase Authentication and Database Guidelines**
- **React and Node.js Documentation**
- **User Interface Design Guidelines**

3. Requirements

3.1 Required States and Modes

- **Landing Page State:** Initial page where users can log in or sign up
    - **Sign-Up:** New users create an account using an email and password
    - **Login:** Returning users log in using their credentials
- **Dashboard State:** After login and account initialization, users are presented with their personalized dashboard
    - **Profile Setup:** Users can update personal information and set preferences
    - **Main Dashboard:** Access to all core features like career questionnaire, resume analysis, and job matching
- **Settings State:** Allows users to adjust account settings, privacy options, and notification preferences

3.2 Functional Requirements

- **FR1:** The website shall support user authentication via email and password
- **FR2:** The website shall provide personalized career path suggestions based on a standardized career questionnaire
- **FR3:** The website shall offer job matching services connected to the user's selected career path
- **FR4:** The website shall provide a resume review feature that highlights areas for improvement
- **FR5:** The website shall include an AI chatbot assistant to answer career-related questions

3.3 External Interface Requirements

- **EIR1:** The website shall integrate with external job boards via APIs to pull job listings
- **EIR2:** The website shall integrate with Firebase for user authentication and data storage

3.4 Internal Data Requirements

- **IDR1:** The website shall store user data securely in a Firebase database
- **IDR2:** The website shall process and store user inputs (e.g., career questionnaire responses) for personalized outputs

3.5 Adaptation Requirements

- **AR1:** The website shall be responsive, adapting the user interface for different screen sizes (e.g., mobile, tablet, desktop)

3.6 Security Requirements

- **SecR1:** The website shall securely store user information, such as email addresses and passwords, using industry-standard encryption
- **SecR2:** The website shall implement secure authentication protocols to prevent unauthorized access

3.7 Environment Requirements

- **ER1:** The website shall be compatible with major web browsers (e.g., Chrome, Firefox, Safari)

3.8 Computer Resource Requirements

- **CRR1:** The website shall be optimized to run smoothly on devices with limited resources, such as mobile phones

3.9 Software Quality Factors

- **SQF1:** The website shall aim for high usability, ensuring that users can easily navigate and use all features
- **SQF2:** The website shall maintain high reliability, with minimal downtime and quick load times

3.10 Design and Implementation Constraints

- **DIC1:** The development of the website is constrained by the capabilities and limitations of the chosen tech stack (React, Node.js, Firebase)

3.11 Personnel-related Requirements

- **PRR1:** The development team shall include at least one full-stack developer proficient in React and Node.js

3.12 Training-related Requirements

- **TRR1:** Users shall not require formal training but will be provided with a tutorial and help documentation within the website

3.13 Other Requirements

- **OR1:** The website shall include accessibility features, ensuring usability for users with disabilities (e.g., screen reader compatibility)

4. Precedence and Criticality of Requirements

- **Security** and **usability** requirements are critical and must be prioritized during development

5. Qualification Provisions

- The website shall undergo usability and security testing to ensure it meets all required standards

6. Requirements Traceability

- A traceability matrix shall be created to ensure each requirement is implemented and tested accordingly

7. Notes

- This document is subject to updates as development progresses and feedback is received

8. Appendices

- **A. glossary**
- **B. Revision History**

9. Evaluation Criteria

- The website's performance, user satisfaction, and adherence to requirements will be evaluated

10. Acronyms and Abbreviations

- **SRS:** Software Requirements Specification
- **UI:** User Interface
- **API:** Application Programming Interface